



Institute of Information Technology

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Lab Report - 03

Lab Title: Linux File Commands (File Management & Directory Management)

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Course Title: Operating System Lab

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1. Objective

In this lab, we practiced frequently used Linux commands for file and directory management, specifically focusing on copying, moving, renaming files and directories, searching within files using grep, and text editing using the Vim editor. The goal was to become comfortable with these powerful Linux terminal operations through hands-on practice.

2. Theory

Linux provides a rich set of command-line tools for managing files and directories. Unlike graphical file managers, the terminal gives precise control over file operations. In this lab, we used the Ubuntu terminal to execute commands for copying, moving, searching, and editing files.

The `cp` command copies files or directories from one location to another. The `mv` command moves or renames files and directories. The `grep` command searches for patterns or text within files and supports options like case-insensitive search, line-number display, and recursive directory search.

Vim is a powerful terminal-based text editor available on most Linux systems. It operates in different modes: normal mode for navigation and commands, and insert mode for typing text. Files can be saved and exited using specific commands in normal mode.

3. Commands Used

Command	Description
<code>cp file1 file2</code>	Copy file1 to file2
<code>cat file1.txt >> file2.txt</code>	Append content of file1 to file2
<code>cp -r dir1 dir2</code>	Recursively copy a directory
<code>mv file1 file2</code>	Rename or move a file
<code>mv file /path/dir/</code>	Move a file to another directory
<code>cp file /path/dir/</code>	Copy a file to another directory
<code>cp file .</code>	Copy a file to the current directory
<code>grep "pattern" file</code>	Search for a pattern in a file
<code>grep -i "pattern" file</code>	Case-insensitive pattern search
<code>grep -n "pattern" file</code>	Show matching lines with line numbers
<code>grep -r "pattern" dir/</code>	Recursively search in all files
<code>grep -v "pattern" file</code>	Show lines NOT containing the pattern
<code>vim filename.txt</code>	Open a file in the Vim text editor

4. Procedure

We opened the terminal on Ubuntu and ran each command one by one. For every command, we first ran it, then observed the output and noted what happened. Below is a step-by-step description of what we did.

4.1 cp file1 file2 - Copy File

Copies the content of file1.txt into file2.txt.

Command:

```
$ touch file1.txt
$ echo "Hello" >> file1.txt
$ cp file1.txt file2.txt
$ cat file2.txt
```

Output:

```
Hello
```

We verified that file2.txt now contains the same content as file1.txt after copying.

4.2 Append File Content

Appends the content of file1.txt to file2.txt without overwriting.

Command:

```
$ echo "World" > file2.txt
$ cat file1.txt >> file2.txt
$ cat file2.txt
```

Output:

```
World
Hello
```

We confirmed that >> appends instead of overwriting, so both 'World' and 'Hello' appeared.

4.3 cp -r dir1 dir2 - Copy Directory

Copies an entire directory recursively from one location to another.

Command:

```
$ cp -r Projects folder1
$ cd folder1
$ ls
```

Output:

```
file1.txt  file2.txt
```

The entire Projects directory and its files were copied into folder1 successfully.

4.4 mv file1 file2 - Rename File

Renames old_name.txt to new_name.txt in the same directory.

Command:

```
$ mv old_name.txt new_name.txt
$ ls
```

Output:

```
file1.txt  file2.txt  new_name.txt
```

The file was renamed and no copy was left behind.

4.5 Move File to Another Directory

Moves new_name.txt to another directory using its absolute path.

Command:

```
$ mv new_name.txt /home/shaon/folder1/
$ cd folder1
$ ls
```

Output:

```
file1.txt  file2.txt  new_name.txt  test.txt
```

The file was successfully moved to folder1 without creating a copy in the original location.

4.6 grep - Search in Files

The grep command searches for text patterns inside files. We created a file called search_file.txt and ran various grep variants on it.

Command:

```
$ echo "This is shaon khan, I am trying my best" > search_file.txt
$ grep "shaon" search_file.txt
```

Output:

```
This is shaon khan, I am trying my best
```

The matching word was highlighted in the terminal output.

Case-insensitive search:

```
$ echo "SHAON Shaon ShAoN Khan" >> search_file.txt
$ grep -i "shaon" search_file.txt
```

Output:

```
This is shaon khan, I am trying my best
SHAON Shaon ShAoN Khan
```

Using -i made the search case-insensitive, so all variations of 'shaon' were matched.

Search with line numbers:

```
$ grep -n "shaon" search_file.txt
```

Output:

```
1:This is shaon khan, I am trying my best
```

The -n flag shows line numbers along with the matching lines.

Recursive search:

```
$ grep -r "main()" codes/
```

Output:

```
codes/test2.txt:main()
```

The -r flag allowed grep to search recursively through all files in the codes/ directory.

Inverse search (lines NOT containing pattern):

```
$ grep -v "debug" file3.txt
```

Output:

```
Line 1: Starting....
```

```
Line 3: Process ....
```

Using -v excluded lines containing 'debug', showing only the non-matching lines.

4.7 Text Editing with Vim

Vim is a terminal-based text editor. We first installed it using sudo, then opened and edited files.

Installation:

```
$ sudo apt install vim -y
```

After getting sudo privileges using the usermod command, vim was installed successfully.

Basic Vim workflow:

- vim file4.txt - Open (or create) a file in Vim
- Press i to enter insert mode and type content
- Press Esc to return to normal (command) mode
- Type :w to save the file
- Type :q to quit Vim
- Type :wq to save and quit at once
- Type :q! to quit without saving

Command:

```
$ touch file4.txt
```

```
$ vim file4.txt
```

Inside Vim, we pressed i and typed the following content:

```
This is the  
first  
line  
everything  
looks  
good
```

```
so
far
The
debug log
```

We then pressed Esc and typed :wq to save and exit.

5. Discussion

This lab covered more advanced file operations compared to the previous lab. A few key observations stood out:

First, the difference between cp and mv is critical. The cp command always leaves the original file intact, while mv removes it from the source location. The -r flag is required when copying entire directories.

Second, the grep command proved very versatile. The -i, -n, -r, and -v flags greatly extend its usefulness. The recursive search with -r is especially powerful when looking for a pattern across an entire project directory.

Third, we encountered permission issues with sudo. The user account was initially not in the sudoers file, so we had to switch to root using su - and run usermod -aG sudo shaon to grant the necessary privileges. After logging back in, sudo commands worked correctly.

Fourth, learning Vim was a significant part of this lab. Vim has a steep learning curve due to its modal editing. We had to remember that typing normally only works in insert mode (press i), and all save/quit commands must be typed in normal mode (press Esc first). Once we understood the modes, the editor became usable.

6. Conclusion

We successfully completed Lab 3 of the Operating System Lab course. In this session, we practiced file copying, directory copying, file renaming, moving files between directories, pattern searching with grep, and text editing using Vim.

Through this lab, we gained practical knowledge of powerful Linux file management commands. We also learned how to handle permission issues by switching to root and granting sudo access. The knowledge of grep and Vim will be especially valuable in future labs and real-world Linux system work.

7. Lab Screenshots

The following screenshots were captured during the lab session and show the actual terminal output for the commands we ran.

```
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
shaon@shaon:~$ mkdir Projects
shaon@shaon:~$ cd Projects
shaon@shaon:~/Projects$ ls
shaon@shaon:~/Projects$ touch file1.txt
shaon@shaon:~/Projects$ echo "Hello" >> file1.txt
shaon@shaon:~/Projects$ cat file1.txt
Hello
shaon@shaon:~/Projects$ cp file1.txt file2.txt
shaon@shaon:~/Projects$ ls
file1.txt  file2.txt
shaon@shaon:~/Projects$ cat file2.txt
Hello
shaon@shaon:~/Projects$ echo "World" > file2.txt
shaon@shaon:~/Projects$ cat file2.txt
World
shaon@shaon:~/Projects$ cat file1.txt >> file2.txt
shaon@shaon:~/Projects$ cat file2.txt
World
Hello
shaon@shaon:~/Projects$ cd
shaon@shaon:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Projects  Public  snap  Templates  Videos
shaon@shaon:~$ cp -r Projects folder1
shaon@shaon:~$ ls
Desktop  Documents  Downloads  folder1  Music  Pictures  Projects  Public  snap  Templates  Videos
shaon@shaon:~$ cd folder1
shaon@shaon:~/folder1$ ls
file1.txt  file2.txt
shaon@shaon:~/folder1$ cat file2.txt
World
Hello
```

```

shaon@shaon:~/Projects$ mv old_name.txt new_name.txt
shaon@shaon:~/Projects$ ls
file1.txt  file2.txt  new_name.txt
shaon@shaon:~/Projects$ cd
shaon@shaon:~$ touch test.txt
shaon@shaon:~$ ls
Desktop  Downloads  Music  Projects  snap  test.txt
Documents  folder1  Pictures  Public  Templates  Videos
shaon@shaon:~$ cp test.txt folder1/
shaon@shaon:~$ cd folder1
shaon@shaon:~/folder1$ ls
file1.txt  file2.txt  test.txt
shaon@shaon:~/folder1$ cd
shaon@shaon:~$ cd Projects
shaon@shaon:~/Projects$ ls
file1.txt  file2.txt  new_name.txt
shaon@shaon:~/Projects$ pwd
/home/shaon/Projects
shaon@shaon:~/Projects$ mv new_name.txt /home/shaon/folder1/
shaon@shaon:~/Projects$ cd folder1
bash: cd: folder1: No such file or directory
shaon@shaon:~/Projects$ cd
shaon@shaon:~$ cd folder1
shaon@shaon:~/folder1$ ls
file1.txt  file2.txt  new_name.txt  test.txt
shaon@shaon:~/folder1$ cd
shaon@shaon:~$ cd Projects
shaon@shaon:~/Projects$ echo "This is shaon khan, I am trying my best" > search_file.txt
shaon@shaon:~/Projects$ ls
file1.txt  file2.txt  search_file.txt
shaon@shaon:~/Projects$ cat search_file.txt
This is shaon khan, I am trying my best
shaon@shaon:~/Projects$ grep "shaon" search_file.txt
This is shaon khan, I am trying my best

```

```

shaon@shaon:~/Projects/codes$ sudo apt install vim -y
[sudo] password for shaon:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  systemd-hwe-hwdb
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  vim-common vim-runtime vim-tiny
Suggested packages:
  ctags vim-doc vim-scripts indent
The following NEW packages will be installed:
  vim vim-runtime
The following packages will be upgraded:

```



"file4.txt" 0L, 0B

```
This is the
first
line
everything
looks
```

Files

far
The
debug log

W

: wq

```
This is the
first
line
everything
looks
good
so
far
The
debug log
```

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